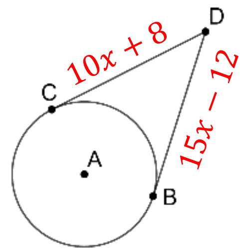
Geometry
Unit 12
Lesson 6 Practice

Name Answer Key____________
Date_____________________
Period____________________

For questions 1 through 3, Circle A is shown. $\overline{CD}$ and $\overline{DB}$ are tangent to the circle.
$CD = 10x + 8$ and $DB = 15x - 12$.

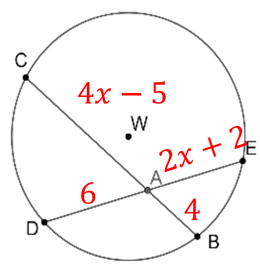


1. What is the value of x?
$10x + 8 = 15x - 12$
$20 = 5x$
$4 = x$

2. What is the length of $\overline{CD}$?
$10(4) + 8 = 40 + 8 = 48$

3. What is the length of $\overline{DB}$?
$15(4) - 12 = 60 - 12 = 48$

For questions 4 through 7, Circle W is shown. Chords $\overline{CB}$ and $\overline{DE}$ intersect at point A.
$CA = 4x - 5$, $AB = 4$, $DA = 6$, and $AE = 2x + 2$.



4. Write an equation you can use to solve for x.
$4(4x - 5) = 6(2x + 2)$
$16x - 20 = 12x + 12$

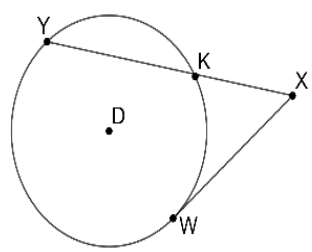
5. Solve the equation you wrote in question 4 for x.
$16x - 20 = 12x + 12$
$4x = 32$
$x = 8$

6. What is the length of $\overline{CB}$?
$4(8) - 5 = 32 - 5 = 27$
$\overline{CB} = 27 + 4 = 31$

7. What is the length of $\overline{DE}$?

$2(8) + 2 = 16 + 2 = 18$
$\overline{DE} = 6 + 18 = 24$

For questions 8 through 10, Circle D is shown. $\overline{XY}$ is a secant to the circle and intersects tangent $\overline{XW}$ at point X. $XW = 10$, $KX = 4$, and $YX = 5x + 10$.
$KX(KX + YK) = (WX)^2$



8. Write an equation that can be used to find the value of x.

$4(5x + 10) = (10)^2$

9. Solve the equation you wrote in question 8 to find the value of x.
$20x + 40 = 100$
$20x = 60$
$x = 3$

10. What is the length of $\overline{YK}$?

$YX = 5(3) + 10 = 15 + 10 = 25$
$25 - 4 = 21 \Longrightarrow YK = 21$